Cloud-mask acceptance audit

OmniCloudMask v4 on Sentinel-2 · Independent model assurance report · 2026-10-26
Prepared by Shane Cairns · All figures computed from public data by the accompanying repository

> **Question.** Before automating on OmniCloudMask, how much cloud and shadow reaches the pixels we accept, how much clear ground do we lose, where does it fail, and which setting should we run?

Summary

- **Use it, with a setting change.** OmniCloudMask is the most accurate mask tested (balanced accuracy 0.942); it leaks 6.1% of accepted pixels by default and 2.0% at the recommended threshold.
- **Stop filtering scenes with the SCL band.** 43.4% of scenes it keeps under a 10% cloud rule are cloudier than that.
- **Expect more leak over water** (10.4%) and missed cloud in mostly clear scenes (16.5% of it).
- **Feed it native-grid, unclipped data.** 30 m imagery resampled to 10 m discards 24.6% of clear ground (native grid: 6.7%); inputs clipped at 0.3 reflectance leak 19.5%.
- **Not sensitive to** processing level (L1C/L2A), per-band calibration drift or NIR misregistration up to 4 px.

Test setup

Data. CloudSEN12+ expert-labeled test split: 965 Sentinel-2 scenes (5.1 km, 10 m) at 193 sites worldwide, 2018-2020, at L1C and L2A. The validation split was used only to choose operating points.

Independence from training data. The model's published training set includes CloudSEN12 material from every split, so its 100,528 training records were checked directly against the test scenes: no shared scene ids (including 2,754 test/validation-tagged training records), no shared acquisitions; one site within 2.6 km of a custom training scene was removed. Removing every scene on a Kappaset training tile leaves the headline unchanged (leak 6.2%).

Comparators. Sentinel-2 QA60 and Sen2Cor SCL (shipped with the products), s2cloudless (upstream defaults), Cloud Score+ (cs ≥ 0.60), UnetMobV2 and SEnSeI v2.

Metrics. *Leak*: share of accepted pixels that are cloud or shadow. *Discard*: share of clear pixels rejected. *Miss*: share of cloud/shadow accepted. Scene filter: keep if estimated cover ≤ 10%. Abstentions count as rejected. 95% confidence intervals by bootstrap over sites; comparisons are paired.

Severity. High: affects a default decision path with large error. Medium: material error in a common condition, or a trade-off the client must set. Low: limited evidence or only under degraded input. Info: checked, no action.

Findings, ranked by severity

#	Severity	Finding and evidence	Recommendation
F1	High	**Coarse imagery resampled to a finer grid is heavily over-masked** Simulated: 30 m data put on a 10 m grid (bilinear, as data cubes do) raises discard to 24.6% (+19.9 pp); the same data run on its native 30 m grid: 6.7% (+2.0 pp). At 20 m: +9.3 pp vs +1.2 pp. Blur alone (1 px) does the same: discard +10.9 pp.	Run the mask on each sensor's native grid, then resample the mask, never the imagery first. Tested: the native run keeps the discard increase near 2 pp.

F2	High	**Scene filters built on the product's own cloud mask keep cloudy scenes** Keeping scenes whose Sen2Cor SCL cloud+shadow cover is at most 10%: 43.4% of kept scenes are actually above 10% (expert labels); SCL under-reports cover by -13 pp on average. The same gate computed from OmniCloudMask: 5.9% above, and it rejects 4.0% of good scenes (SCL: 16.5%).	Gate scenes on OmniCloudMask-derived cover, not the SCL band. Tested: cloud reaching the pipeline inside kept scenes falls from 3.3% to 1.6% [0.7, 2.7] after masking.
F3	Medium	**At its default setting, about 6% of accepted pixels are cloud or shadow** Leak 6.1% [5.3, 7.0] of accepted pixels; thin cloud 2.9%, shadow 1.8%, thick 1.3%. Discard 4.6% [4.0, 5.2] of clear pixels.	For leak-sensitive uses (time series, change detection), accept a pixel only if P(clear) >= 0.85 (chosen on the validation split): test leak 2.0% [1.4, 2.6] at 13.6% discard. Do not buffer the mask instead: an 8-pixel buffer still leaks 3.6% at 13.1% discard.
F4	Medium	**Clipped 8-bit inputs let bright cloud through** Simulated lossless 8-bit with reflectance clipped at 0.3 (close to the Sentinel-2 true-colour stretch): leak 19.5% (+13.4 pp); clip at 0.6: +2.4 pp; at 1.0: +0.5 pp. JPEG compression adds only +0.8 pp at quality 95 and +4.0 pp at quality 50 on top of clipping.	Feed unclipped reflectance (12/16-bit), or 8-bit only with the range kept to at least 1.0 reflectance; do not run the mask on visual/ stretched products.
F5	Medium	**Cloud over water leaks more often (10.4% vs 6.1% overall)** Leak over water 10.4% [6.4, 14.7] across 99 sites (thick cloud alone 4.0%, vs 0.9% over grassland); the two worst sites hold 33% of it, so it is not one bad scene.	For coastal, inland-water and flood work, use the stricter threshold above and spot-check a local sample before relying on the mask.
F6	Medium	**Small amounts of cloud are missed in mostly clear scenes** In scenes with 1-30% cloud, 16.5% [13.6, 19.6] of the cloud and shadow present is accepted as clear (vs 3.3% in overcast scenes). These are the scenes a cover gate keeps.	Covered by the threshold recommendation; the tested buffer does not fix it as cheaply.
F7	Medium	**Usable ground is discarded in heavily clouded scenes** In scenes with 70-99% cloud, 19.8% [17.3, 22.5] of the clear pixels are rejected.	If gap-filling from cloudy scenes matters, validate a lower threshold for that regime (not tested here).
F8	Low	**Snow and ice: higher leak, but weak evidence** Leak 13.0% [6.4, 18.4], only 14 sites, 67% of it from two sites and mostly thin cloud over snow, where expert labels are least certain.	Validate locally before using the mask in snow-covered regions.
F9	Low	**Noisier sensors leak more** Simulated Gaussian noise of 0.005 reflectance (several times Sentinel-2's) raises leak by +2.7 pp; 0.01 by +6.8 pp.	Re-validate before using the mask on lower-SNR small-satellite imagery.
F10	Info	**Verified non-issues** L1C vs L2A input: leak +0.2 pp [0.0, 0.4]. Per-band calibration drift: 99.99% pixel agreement with the clean run (the model normalizes each band per image). NIR misregistration up to 4 px: at most +0.9 pp on leak or discard. Results are stable across continents, seasons and sun elevation (overlapping CIs).	No action.

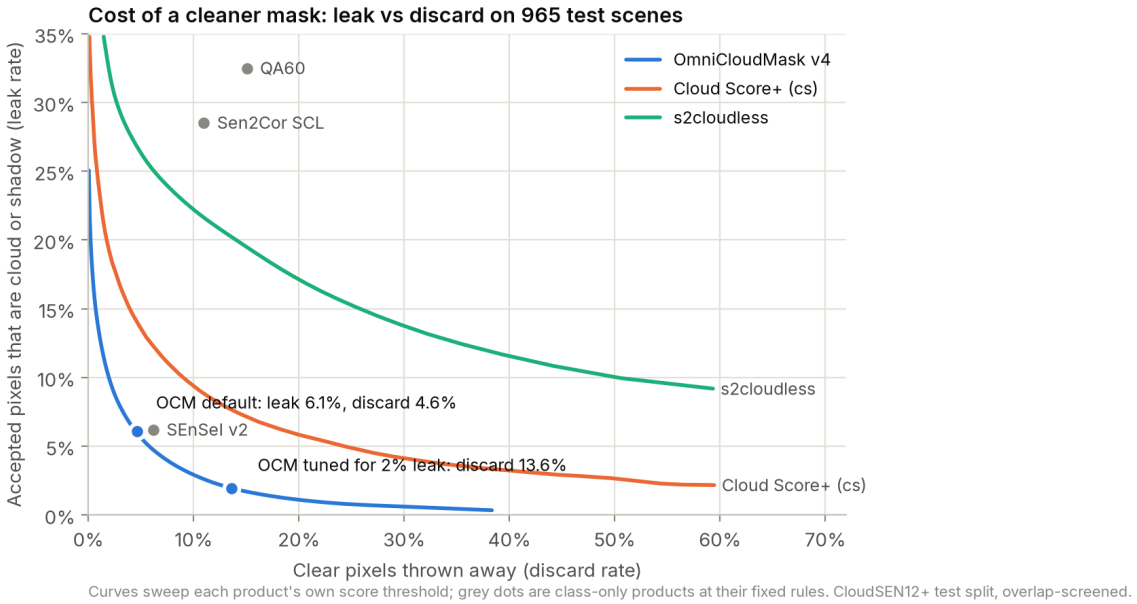


Figure 1. Leak against discard for every threshold of the score-based masks; grey points are fixed-rule products. Test split.

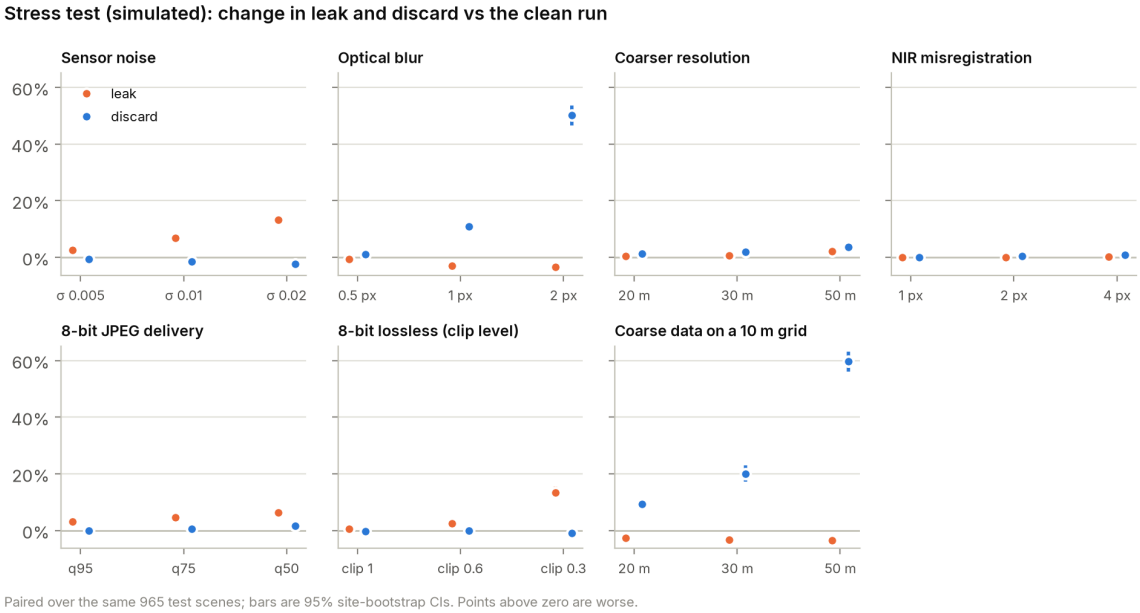


Figure 2. Stress test (simulated degradations): change in leak and discard against the clean run on the same scenes, with 95% site-bootstrap CIs. "Coarse data on a 10 m grid" is 20/30/50 m data resampled to 10 m before masking; "Coarser resolution" is the same data masked on its native grid.

Recommendations

1. Run the mask on each sensor's native grid; resample the mask, not the imagery.
2. Gate scenes on OmniCloudMask cover instead of the Sen2Cor SCL band.
3. For leak-sensitive uses, accept pixels only at P(clear) >= 0.85; do not rely on buffering.
4. Feed unclipped reflectance; never run the mask on stretched 8-bit products.
5. Spot-check a local sample before relying on the mask over water.
6. If you gap-fill from cloudy scenes, validate a lower threshold for them.

Limitations

- Expert labels are least certain for thin cloud over snow and at cloud edges; part of the measured leak there is label disagreement.
- The test split is cloudier than a typical archive; per-regime results allow reweighting to a client's own cloud climatology.
- Sentinel-2 only; the model's sensor-agnostic claim was not tested. Degradations are simple simulations that bound sensitivity.

Reproducibility: pinned dataset revision, model weights verified by SHA-256, one-command pipeline (`make all`), unit tests. Data: CloudSEN12+ (CC0); contains modified Copernicus Sentinel data 2018-2020; ESA WorldCover (CC BY 4.0). Model: OmniCloudMask (MIT).